# Curriculum Vitae - Sandesh Kalantre

## Contact Information

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## Interests

$\mathcal{E}$xperimental Condensed Matter Physics, Quantum Computers,
Quantum Transport, Machine Learning

## Education

Indian Institute of Technology, Bombay, India — July 2014 - April 2018
Bachelor of Technology (B.Tech) in Engineering Physics, Minor in Computer Science
GPA : 9.88/10
**Ranked $1^{st}/42$ in the Physics Department and $2^{nd}/880$ in the Institute**

## Publications and Preprints

**S. S. Kalantre**, J. P. Zwolak, S. Ragole, X. Wu, N. Zimmerman, M. D. Stewart and J. M. Taylor, *Machine learning techniques for state recognition and auto-tuning in quantum dots*, arXiv:1712:04914

## Research Presentations

- *ML and Quantum Dots: Generation of Training Datasets and Autotuning*, APS March Meeting, Los Angeles, CA, USA (March 2018)
- *ML and Quantum Dots: Navigation in Quantum Dot Land*, Quantum Seminar, NIST, Gaithersburg, MD, USA (July 2017)
- *ML and Quantum Dots: Navigation in Quantum Dot Land*, seminar at University of Maryland-College Park, MD, USA (July 2017)
- *Spin Qubits in Semiconductors*, Poster for ISTernship Program, IST Austria, Vienna (July 2016)

## Honours and Awards

- Institute Academic Award for highest GPA in the Physics Department, IIT Bombay (2015-2017)
- **Silver Medal** at $7^{th}$ International Olympiad on Astronomy and Astrophysics, Greece (2013)
- **Gold Medal** at $6^{th}$ International Olympiad on Astronomy and Astrophysics, Brazil (2012)
- **Gold Medal** at $16^{th}$ International Astronomy Olympiad, Kazakhstan (2011) and a special diploma for **ranking** $1^{st}$ among participants from 21 countries

## Scholarships & Fellowships

**S. N. Bose Scholars Program** — 2017
$2000 travel support for summer-research experience at leading US institutions

**ISTernship - Institute of Science and Technology (IST) Austria** — 2016
€2700 award to pursue research as part of the ISTernship (awarded to around 40 students worldwide)

**National Initiative on Undergraduate Science (NIUS) Program in Physics** — 2015
Financial support for summer at leading institutions in India (awarded to 20 students in the country)

**Kishore Vigyan Protsahan Yojana (KVPY)** — 2013
Financial awarded for promotion of basic sciences among high school students (awarded to the top 250 students in the country)

**National Talent Search Scholarship (NTSE)** — 2010
Awarded by the National Council for Educational Research and Training, Govt. of India to the top 1000 high school students in the country as support for secondary school education.

## Key Research Experience

National Institute of Standards and Technology, Gaithersburg, MD, USA
**Machine Learning Techniques for auto-tuning of Quantum Dots**
*Advisor: Dr. Jacob M. Taylor* *Summer 2017*

- Simulated a Thomas-Fermi based model for capacitance matrix calculations in linear arrays of quantum dots
- Developed novel machine learning techniques with deep and convolutional neural networks to characterize charge configurations of semiconductor quantum dots
- Introduced a procedure of "auto-tuning" to automatically achieve a desired configuration of a quantum dot array
- Generated a dataset of simulated single and double quantum dots using the supercomputing cluster at University of Maryland as a resource for future machine learning studies

Institute of Science and Technology, Vienna, Austria
**Spin Qubits in Semiconductor (Ge) Nanostructures**
*Advisor: Dr. Giorgos Katsaros* *Summer 2016*

- Studied theoretical aspects of transport in quantum dot and nanowire systems in semi-conductors, and their potential application as spin qubits
- Worked on measurement of conductance to observe *Coulomb blockade and Diamond Plots* in p-type transistors at 4 K (Liquid He)
- Explored the theory behind *superconducting resonators* to couple double-quantum dot systems and tried measurement of resonance in a Nb resonator at 4 K
- Extensively worked with **QTLab** environment for device control and developed drivers for the Arbitrary Waveform Generator (AWG), Vector Network Analyser (VNA) and Signal Generator

Indian Institute of Technology, Bombay, India
**Quantum Dot Fabrication in Si/SiGe and ESR Selectivity of Donor spin qubits in a slanting Zeeman field (Senior Thesis)**
*Advisor: Prof. Suddhasatta Mahapatra* *2017-2018 (expected)*

- Working in a nano-fabrication facility to pattern nano-scale devices in semiconductor heterostructures
- Gathered experience on usage of tools like e-beam metal evaporation system for metal deposition, Scanning Electron Microscopy (SEM) and Electron Beam Lithography (EBL)
- Worked on a theoretical problem of Electron Spin Resonance (ESR) selectivity of Phosphorous Donor qubits in micromagnet fields
- Proposed micromagnet architectures that can be used to achieve selectivity while not compromising scalability for large linear arrays of donor qubits (manuscript in preparation)

## Other Research Experience

Besides working on Experimental and Theoretical Quantum Computation, I have also explored certain aspects of Theoretical Condensed Matter Physics, Astrophysics, Non-Linear dynamics and worked with devices such as FPGAs, micro-controllers (Arduino) and Raspberry Pi.

**Non-equilibrium Green's function calculation of transport in superconducting-normal junction**
*Advisor: Prof. Bhaskaran Muralidharan, Indian Institute of Technology, Bombay, India* *2016-2017*

- Employed non-equilibrium Green's function formalism to model transport in quantum systems
- Determined the effects of angular momentum subbands and Andreev Bound states to the current-phase relationship in nanowires connected to superconducting contacts

**Quantum Algorithms and Cryptography**
*Advisor: Prof. Prasanta Panigrahi, IISER Kolkata* *Summer 2015*

- Learned about Bell's Theorem & CHSH inequality and Quantum Key Distribution schemes
- Applied Grover's algorithm to search for the existence of an Eulerian circuit in a undirected graph given the adjacency list representation [Report]

**Quantum Entanglement and Teleportation**
*Prof. Bhaskaran Muralidharan, Indian Institute of Technology, Bombay, India* *Autumn 2015*

- Investigated the exchange Hamiltonian among electronic spins and nuclear spin baths
- Studied the use of Bell measurements of NV electronic spin and $N$ nuclear spin to accomplish teleportation of quantum states

**Quantum Hall Effect**
*Reading Project, Prof. Vikram Rentala, Department of Physics, IIT Bombay* Autumn 2016

- Investigated various approaches to the derivation of quantized Hall conductivity arising in the integer quantum Hall effect
- Explored the role of Topology and Chern number in condensed matter systems

**Pair-production anomaly from propagation of VHE gamma-rays**
*Prof. Vikram Rentala, Department of Physics, IIT Bombay* *Autumn 2015*

- Investigated various components of Extra-galactic Background Light (EBL) at $\mu m$ wavelengths and their relative contributions to the pair production process
- Worked on Inter-Halo Light (light from stars outside galaxies) models to scrutinize its effect on VHE spectral indices and the spectral-break [Report]

**B-mode spectrum and Inflation Models**
*National Initiative for Undergraduate Science (NIUS) - Astronomy* *December 2015*

- Read introductory literature on the need for inflation in the early universe and slow-roll equations for the inflation field
- Analyzed different inflation models and used CAMB code to obtain B-mode spectrum which was compared with Planck-BICEP data. [Report]

## Course & Seminar Projects

**Twisted Light** *PH440 - Atomic and Molecular Physics*
I read the basics of orbital angular momentum (OAM) of light and presented a literature survey on OAM experiments in the last 20 years. [Slides]

**Jarzynski's Equality** *EP413 - Advanced Statistical Physics*
We studied a novel result in non-equilibrium thermodynamics, Jarzynski's equality, relating non-equilibrium work and the free energy difference. [Slides]

**Green's functions in Physics** *PH308 - Introduction to Condensed Matter Physics*
We performed an extensive survey on the application of Green's functions to problems in theoretical condensed matter physics. [Slides, Poster]

**Webcam Spectrograph using Raspberry Pi (RPi)** *EP226 - Waves and Oscillations*
We used CD as a diffraction grating & the camera of a RPi to construct a portable spectrograph.

**Three Body Simulation using FPGA** *EP230 - Digital Electronics Lab*
We Programmed an Altera De0-Nano board using VHDL to perform real time simulation of 3 bodies interacting gravitationally.

**NumCpp - a mathematical library in C++** *CS101 - Computer Programming and Utilization*
We wrote a syntax parser and an arbitrary precision mathematical library in C++

## Workshops

**Pulsar Observatory for Students (POS) - 2014**
*Radio Astronomy Center, Ooty, India*
Studied the operation of the Ooty Radio Telescope (ORT), a 530-metre (1,740 ft) long and 30-metre (98 ft) wide Cylindrical Paraboloid telescope. Collected data for pulsars using the ORT and analzsed it to explore various properties such as the Dispersion Measure (DM), modulation index and pulse broadening due to interstellar scattering.

## Mentoring Experience

**Teaching Assistant**
*For Quantum Physics, Electromagnetism, Partial Differential equations courses* *2015-2017*

- Mentored a batch of around 50 students in the course content and involved in the evaluation of exams and quizzes

**Institute Student Mentor** *Indian Institute of Technology, Bombay* *2017*

- Mentor to a wing of 12 freshmen to assist them in settling to college life and the academic setting of an university
- Involved in conducting basic computer learning classes for the computer illiterate students to help them cope with computer programming courses

**Resource Person and Student Facilitator**
*Indian Astronomy Olympiad Program, HBCSE - TIFR, Mumbai* *2015*

- Mentor to about top 30 high school students at the selection camp for the Indian Team to the International Olympiad on Astronomy and Astrophysics (IOAA)
- The team was awarded 3 Gold & 2 Silver medals in the International Olympiad on Astronomy and Astrophysics (IOAA), Indonesia, 2015, which was *India's best result in 9 years.*

## Positions of Responsibility

**Manager - Maths and Physics Club, IIT Bombay** 2016-2017
*Students' Technical Activities Body (STAB), IIT Bombay*

- Led a team of 6 sophomore conveners in organization of events fostering to the enthusiasm of students in Physics and Mathematics; catering to around **400-500** students on campus and having an outreach of around **6000** online
- Introduced a new initiative in Summer 2016, *Summer of Science* in which students studied varied topics such as Game Theory, Cosmology and Quantum Computation under senior student mentors

**General Secretary** *Department of Physics, IIT Bombay* *2017-2018*

- As the top student representative of the physics department's bachelor and masters' students, I co-ordinate and communicate students issues to the faculty and other representatives at the institute level.
- Responsible for the organization of departmental events such as the valedictory function, degree convocation and freshmen introduction sessions.

**Academic Resource Person & IPho-rum developer** 2015
*Browser Application for International Physics Olympiad (IPhO), Mumbai, 2015*

- Worked in a team of 2 in development of IPho-rum — a browser based application for tasks such as voting, translation upload and feedback submission among *approx 100 users*; used *Node.js* as the server application and Mongodb as the primary database
- Involved as a student grader in the academic team

## Popular Talks

**A Brief Introduction to Quantum Computation**
*Maths and Physics Club, IIT Bombay* *March 2018*
Starting with the history of computing, I presented how quantum computing emerges as the natural 'next step'. I talked about killer-apps such as integer factorization, quantum many-body simulation.

**Cosmology: Dark Things on a White Background**
*Kritikka - The Astronomy Club, IIT Bombay* *March 2018*
Gave an elementary introduction to cosmological models using Newtonian mechanics including the necessity of dark matter and dark energy to explain observations.

**Celestial Mechanics**
*Kritikka - The Astronomy Club, IIT Bombay* *September 2015,2016*
Presented a complete derivation of the dynamics of the two-body problem using Newtonian mechanics. Talked about chaos in dynamics systems in relation on the three and N body problems.

## Additional Courses

**Physics** : Relativistic Quantum Mechanics, Applied Solid State Physics, Advanced Statistical Physics, Group Theory, Theoretical Condensed Matter Physics, Quantum Computation & Information, General Relativity, Non-Linear Dynamics

**Others** : Category Theory & Hyperplane arrangements, Design and Analysis of Algorithms, Operating Systems, Data Structures and Algorithms, Signals and Systems, Computer Networks, Microprocessors Lab, Shakespearean Afterlives

## Technical Skills

**Programming** : C++, Python

**Scientific Packages** : TensorFlow, Synopsys - TCAD, NumPy, SciPy, Matplotlib, QuTip, Astropy, gnuplot, MATLAB, Mathematica, Octave, Origin, GNU MPFR library

**Softwares** : LaTeX, git, InkScape, node.js, Wireshark, PyGTK

## Extra-curricular activities

- Won a Bronze medal in Astronomy hackathon as part of InterIIT Tech Meet, IIT Madras, December 2017.
- Interested in all things in print, from abstract mathematics, history & mythology to Shakespeare & Whitman
- Built a Kelvin Water Dropper and demonstrated build-up of a potential difference under Maths and Physics club, IIT Bombay
- Secured first place in *Online Physics Brawl 2016* in the open category
- Associated with National Service Scheme as a volunteer at NGOs for underprivileged children in Powai, Mumbai
- Secured first prize in Debate competition held at Dyanamata High School, Maharashtra

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